

Systolic Array (Matrix Multiplication Accelerator)

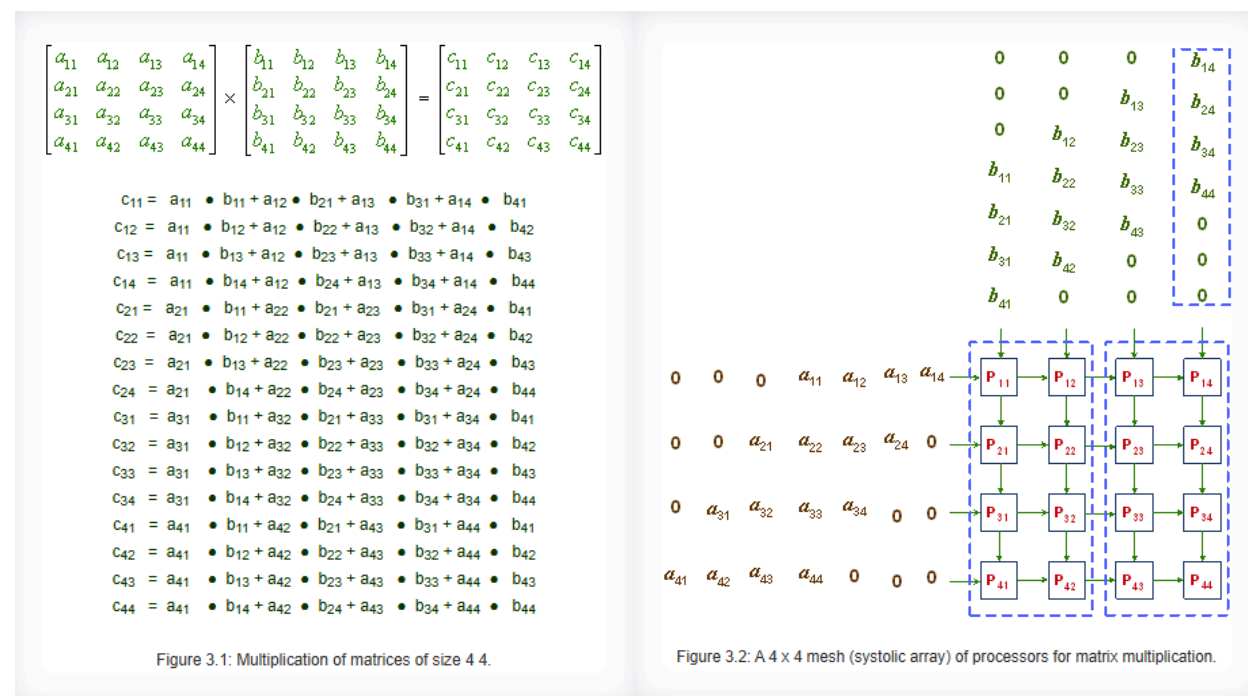
Spring 2025

As an onboarding freshmen member of Digital Design at Cornell Custom Silicon Systems, I independently designed, implemented, verified, synthesized, and documented a systolic array for accelerating matrix multiplication. The project kickstarted the team's development of a RISC-V machine learning processor architecture IP starting Fall 2025.

Background

Matrix multiplication is ubiquitous in machine and deep learning tasks, and systolic arrays (used in Google's TPUs) can accelerate this critical operation for two-dimensional input and weight matrices.

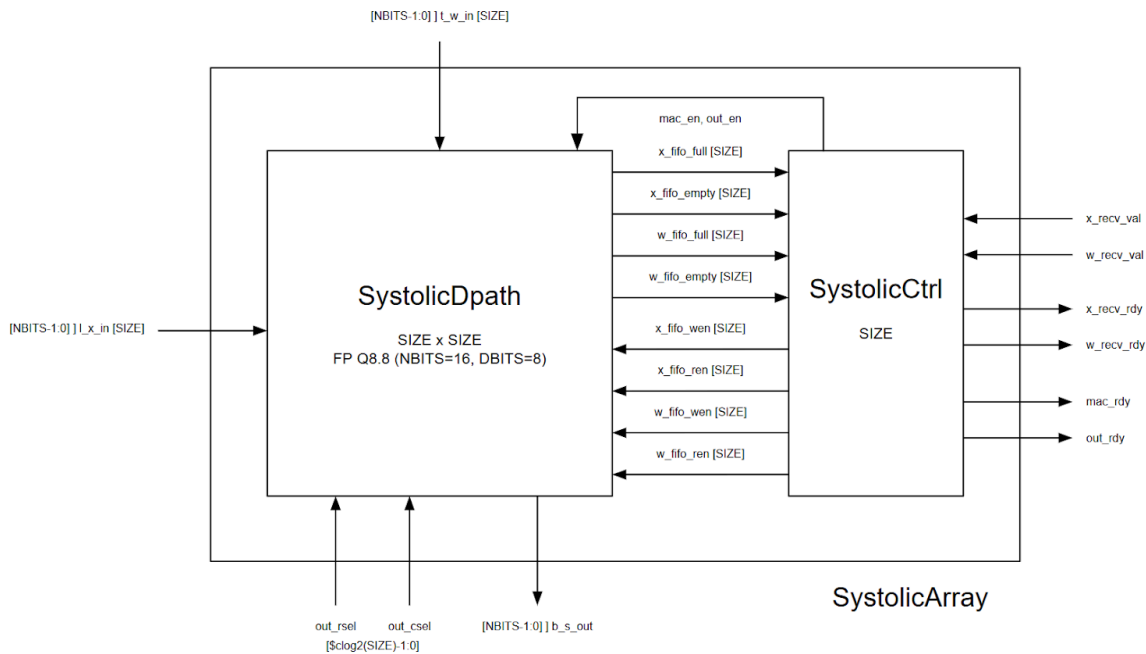
Systolic arrays typically consist of $N \times N$ processing elements (PE) where each PE not only performs the typical multiply-accumulate (MAC) operation but also propagates inputs, weights, and/or summations to other PEs such that every value in the matrices is constantly busy.



The two figures above show how data flows through a systolic array. Note that this shows an output-stationary data flow (used in our systolic array) where each PE has its own MAC output and propagates the inputs and weights to other PEs, and there are several other data flow types (e.g., input-stationary, weight-stationary, etc).

RTL Design

Top



Our systolic array uses the output-stationary data flow to multiply two 2D matrices with signed fixed point (FP) values. The following are top-level parameters and I/Os.

Parameters

- **SIZE**: number of PEs per row and per column (default: 4)
- **NBITS**: FP bitwidth of input and weight values (default: 16)
- **DBITS**: number of decimal bits of input and weight values (default: 8)

Inputs

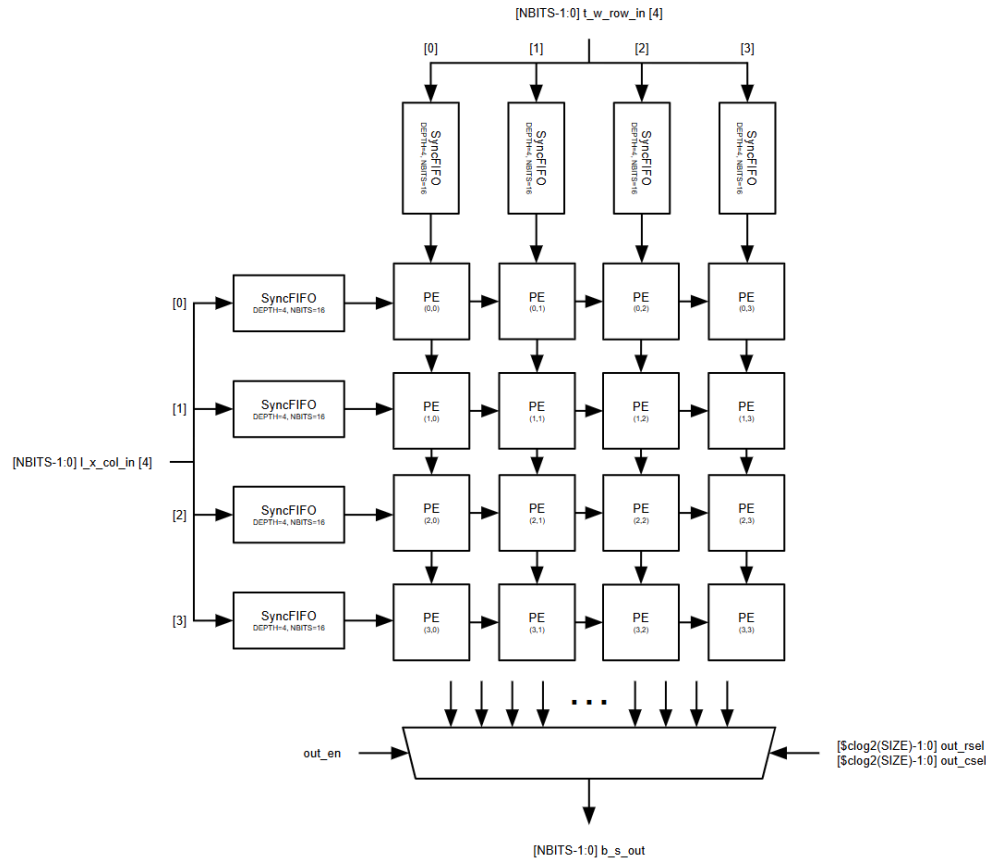
- **clk**: clock signal
- **rst**: reset signal
- $[NBITS-1:0] l_x_col_in [SIZE]$: a column of the input matrix
 - Zero-pad if necessary
 - Deserializer outside of the systolic array may be helpful
- **x_recv_val**: signal for indicating that an input column is being driven into the systolic array
- $[NBITS-1:0] t_w_row_in [SIZE]$: a row of the weight matrix
 - Zero-pad if necessary
 - Deserializer outside of the systolic array may be helpful
- **w_recv_val**: signal for indicating that a weight row is being driven into the systolic array
- $[\$clog2(SIZE)-1:0] out_rsel$: row index of output matrix to read

- $[\$clog2(SIZE)-1:0]$ out_csel: column index of output matrix to read

Outputs

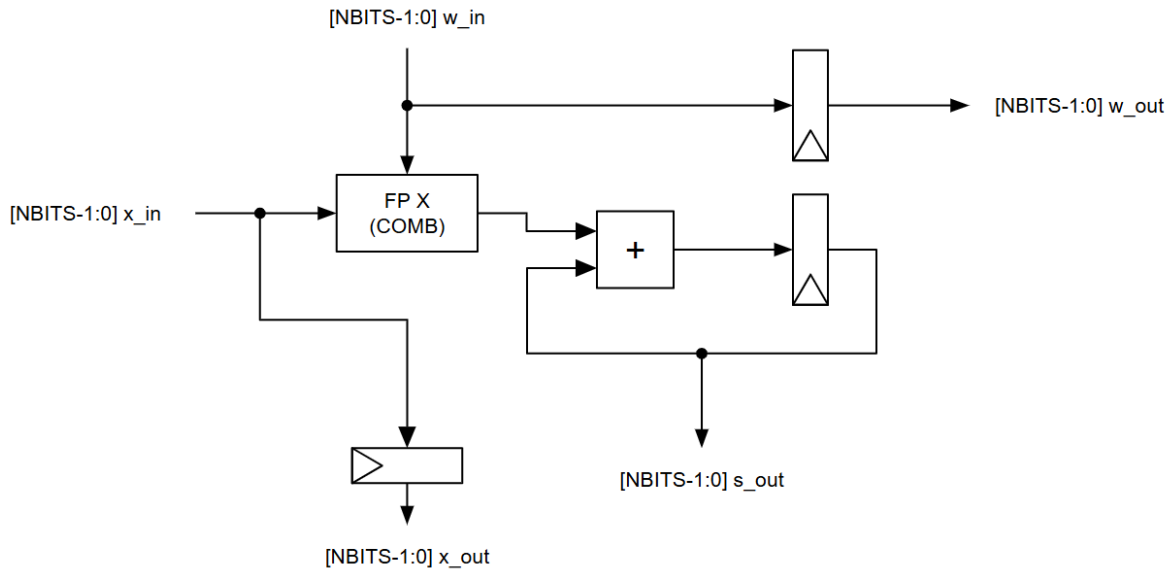
- x_recv_rdy: signal indicating that an input matrix is fully loaded in the systolic array
- w_recv_rdy: signal indicating that a weight matrix is fully loaded in the systolic array
- mac_rdy: signal indicating that the systolic array is performing MAC operations
- out_rdy: signal indicating that all outputs in the systolic array are stable
- $[NBITS-1:0]$ b_s_out: output matrix value at the specified index

Data Path

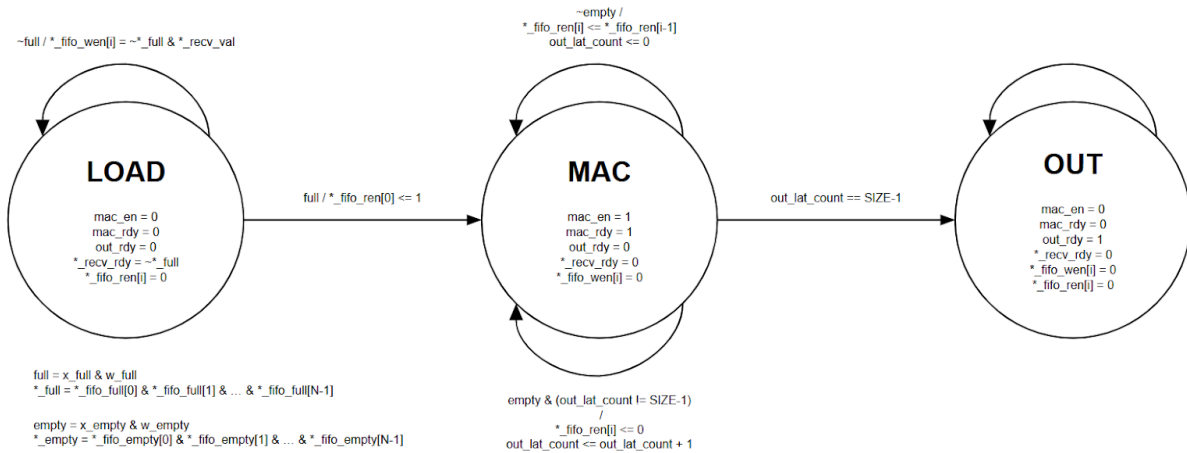


For an $N \times N$ sized systolic array, there are $N \times N$ PEs that perform MAC operations and N synchronous FIFO buffers each for the input and weight matrices to store the matrices and flow the matrices into the array of PEs.

Each PE takes in one input and weight value. These values are driven into a combinational FP multiplier, and the product is accumulated to a sum register (i.e., the PE's output). The input and weight entering each PE is also outputted through a register and delivered to adjacent PEs.



Control Logic



LOAD

- Load one column of the input matrix per cycle by setting $x_recv_val = 1$
 - Input matrix can be loaded in descending or ascending order (must match weight matrix's loading order) of columns
 - Zero-pad the input matrix if necessary
 - $x_recv_rdy = 0$ indicates the input matrix FIFOs are full (cannot load)
 - $x_recv_rdy = 1$ indicates the input matrix FIFOs are not full (can load)
- Load one row of the weight matrix per cycle by setting $w_recv_val = 1$
 - Weight matrix can be loaded in descending or ascending order (must match input matrix's loading order) of rows

- Zero-pad the input matrix if necessary
- `w_recv_rdy = 0` indicates the weight matrix FIFOs are full (cannot load)
- `w_recv_rdy = 1` indicates the weight matrix FIFOs are not full (can load)

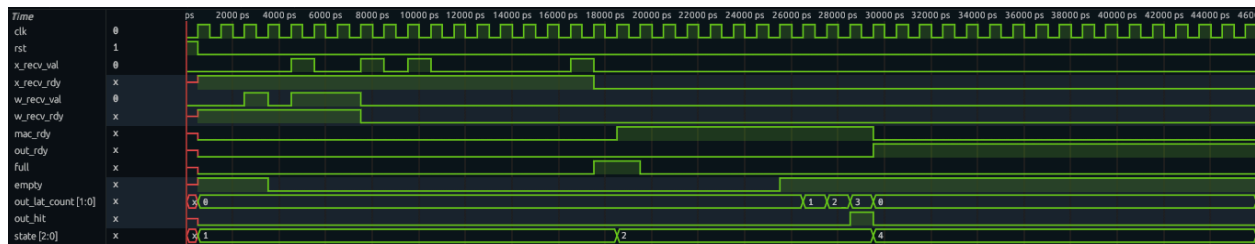
MAC

- Once the input and weight FIFOs are full, the PEs are enabled to perform MAC operations (`mac_rdy = 1, out_rdy = 0`)
- During MAC, the input and weight FIFOs cannot be written

OUT

- The systolic array enters OUT state when the last PE finishes its computation and all outputs become stable (`mac_rdy = 0, out_rdy = 1`)
- Setting the output row (`out_rsel`) and column (`out_csel`) will combinationally dump the selected PE's output (`b_s_out`)
 - Attempting to read outputs during non-OUT states will return all zeros (`out_rdy` sets `out_en` of data path)
- During OUT, the input and weight FIFOs cannot be written
- Reset the systolic array to clear the PEs and start a new matrix multiplication (returns to LOAD)

The following waveform corresponds to an example execution of a 4x4 systolic array.



Design Verification

Cocotb (Synopsys VCS) tests for the systolic array's top-level consist of:

- Directed loading of an input matrix
- Directed loading of a weight matrix
- Directed synchronous loading of input and weight matrices
- Randomized asynchronous loading input and weight matrices
- Directed synchronous loading of input and weight matrices and full LOAD/MAC/OUT sequence
- Randomized asynchronous loading of input and weight matrices and full LOAD/MAC/OUT sequence

The following coverage report is from running all Cocotb tests on a 4x4 array (FP Q8.8).

Top

Module : SystolicArray

SCORE	LINE	TOGGLE
100.00		100.00

Source File(s) :
/home/js2992/c2s2/c2s2_ip/src/systolic/SystolicArray.v

Module self-instances :

NAME	SCORE	LINE	TOGGLE
SystolicArray	100.00		100.00

Module Instance : SystolicArray

Instance :

SCORE	LINE	TOGGLE
100.00		100.00

Instance's subtree :

SCORE	LINE	TOGGLE
99.31	98.86	99.75

Parent :
none

Subtrees :

NAME	SCORE	LINE	TOGGLE
ctrl	96.88	93.75	100.00
dpath	99.87	100.00	99.75

Since this is the module's only instance, the coverage report is the same as for the module.

Toggle)

Toggle Coverage for Module : SystolicArray

	Total	Covered	Percent
Totals	13	13	100.00
Total Bits	60	60	100.00
Total Bits 0->1	30	30	100.00
Total Bits 1->0	30	30	100.00
Ports	11	11	100.00
Port Bits	56	56	100.00
Port Bits 0->1	28	28	100.00
Port Bits 1->0	28	28	100.00
Signals	2	2	100.00
Signal Bits	4	4	100.00
Signal Bits 0->1	2	2	100.00
Signal Bits 1->0	2	2	100.00

Port Details

Name	Toggle	Toggle 1->0	Toggle 0->1	Direction
clk	Yes	Yes	Yes	INPUT
rst	Yes	Yes	Yes	INPUT
x_recv_val	Yes	Yes	Yes	INPUT
x_recv_rdy	Yes	Yes	Yes	OUTPUT
w_recv_val	Yes	Yes	Yes	INPUT
w_recv_rdy	Yes	Yes	Yes	OUTPUT
mac_rdy	Yes	Yes	Yes	OUTPUT
out_rdy	Yes	Yes	Yes	OUTPUT
out_rsel[1:0]	Yes	Yes	Yes	INPUT
out_csel[1:0]	Yes	Yes	Yes	INPUT
b_s_out[15:0]	Yes	Yes	Yes	OUTPUT

Signal Details

Name	Toggle	Toggle 1->0	Toggle 0->1
mac_en	Yes	Yes	Yes
out_en	Yes	Yes	Yes

Data Path

Module : SystolicDpath

SCORE	LINE	TOGGLE
100.00		100.00

Source File(s) :
/home/js2992/c2s2/c2s2_ip/src/systolic/SystolicDpath.v

Module self-instances :

NAME	SCORE	LINE	TOGGLE
SystolicArray.dpath	100.00		100.00

Module Instance : SystolicArray.dpath

Instance :

SCORE	LINE	TOGGLE
100.00		100.00

Instance's subtree :

SCORE	LINE	TOGGLE
99.87	100.00	99.75

Parent :

SCORE	LINE	TOGGLE	NAME
100.00		100.00	SystolicArray

Subtrees :

NAME	SCORE	LINE	TOGGLE
g_pe_rows[0].g_pe_cols[0].PE	99.86	100.00	99.72
g_pe_rows[0].g_pe_cols[1].PE	99.86	100.00	99.72
g_pe_rows[0].g_pe_cols[2].PE	99.86	100.00	99.72
g_pe_rows[0].g_pe_cols[3].PE	99.86	100.00	99.72
g_pe_rows[1].g_pe_cols[0].PE	99.86	100.00	99.72
g_pe_rows[1].g_pe_cols[1].PE	99.86	100.00	99.72
g_pe_rows[1].g_pe_cols[2].PE	99.86	100.00	99.72
g_pe_rows[1].g_pe_cols[3].PE	99.86	100.00	99.72
g_pe_rows[2].g_pe_cols[0].PE	99.86	100.00	99.72
g_pe_rows[2].g_pe_cols[1].PE	99.86	100.00	99.72
g_pe_rows[2].g_pe_cols[2].PE	99.86	100.00	99.72
g_pe_rows[2].g_pe_cols[3].PE	99.86	100.00	99.72
g_pe_rows[3].g_pe_cols[0].PE	99.86	100.00	99.72
g_pe_rows[3].g_pe_cols[1].PE	99.86	100.00	99.72
g_pe_rows[3].g_pe_cols[2].PE	99.86	100.00	99.72
g_pe_rows[3].g_pe_cols[3].PE	99.86	100.00	99.72
g_tensor_fifo[0].TensorFIFO	100.00	100.00	100.00
g_tensor_fifo[1].TensorFIFO	100.00	100.00	100.00
g_tensor_fifo[2].TensorFIFO	100.00	100.00	100.00
g_tensor_fifo[3].TensorFIFO	100.00	100.00	100.00
g_weight_fifo[0].WeightFIFO	100.00	100.00	100.00
g_weight_fifo[1].WeightFIFO	100.00	100.00	100.00
g_weight_fifo[2].WeightFIFO	100.00	100.00	100.00
g_weight_fifo[3].WeightFIFO	100.00	100.00	100.00

Since this is the module's only instance, the coverage report is the same as for the module.

Toggle)

Toggle Coverage for Module : SystolicDpath

	Total	Covered	Percent
Totals	7	7	100.00
Total Bits	48	48	100.00
Total Bits 0->1	24	24	100.00
Total Bits 1->0	24	24	100.00
Ports	7	7	100.00
Port Bits	48	48	100.00
Port Bits 0->1	24	24	100.00
Port Bits 1->0	24	24	100.00

Port Details

Name	Toggle	Toggle 1->0	Toggle 0->1	Direction
clk	Yes	Yes	Yes	INPUT
rst	Yes	Yes	Yes	INPUT
mac_en	Yes	Yes	Yes	INPUT
out_en	Yes	Yes	Yes	INPUT
out_rsel[1:0]	Yes	Yes	Yes	INPUT
out_csel[1:0]	Yes	Yes	Yes	INPUT
b_s_out[15:0]	Yes	Yes	Yes	OUTPUT

Control Logic

Module : SystolicCtrl

SCORE	LINE	TOGGLE
96.88	93.75	100.00

Source File(s) :

/home/js2992/c2s2/c2s2_ip/src/systolic/SystolicCtrl.v

Module self-instances :

NAME	SCORE	LINE	TOGGLE
SystolicArray.ctrl	96.88	93.75	100.00

Module Instance : SystolicArray.ctrl

Instance :

SCORE	LINE	TOGGLE
96.88	93.75	100.00

Instance's subtree :

SCORE	LINE	TOGGLE
96.88	93.75	100.00

Parent :

SCORE	LINE	TOGGLE	NAME
100.00		100.00	SystolicArray

Subtrees :

NAME	SCORE	LINE	TOGGLE
no children			

Since this is the module's only instance, the coverage report is the same as for the module.

Line Toggle

	Total	Covered	Percent
Totals	16	16	100.00
Total Bits	38	38	100.00
Total Bits 0->1	19	19	100.00
Total Bits 1->0	19	19	100.00
Ports	9	9	100.00
Port Bits	18	18	100.00
Port Bits 0->1	9	9	100.00
Port Bits 1->0	9	9	100.00
Signals	7	7	100.00
Signal Bits	20	20	100.00
Signal Bits 0->1	10	10	100.00
Signal Bits 1->0	10	10	100.00

Port Details

Name	Toggle	Toggle 0->1	Toggle 1->0	Direction
clk	Yes	Yes	Yes	INPUT
rst	Yes	Yes	Yes	INPUT
mac_en	Yes	Yes	Yes	OUTPUT
x_recv_val	Yes	Yes	Yes	INPUT
x_recv_rdy	Yes	Yes	Yes	OUTPUT
w_recv_val	Yes	Yes	Yes	INPUT
w_recv_rdy	Yes	Yes	Yes	OUTPUT
mac_rdy	Yes	Yes	Yes	OUTPUT
out_rdy	Yes	Yes	Yes	OUTPUT

Signal Details

Name	Toggle	Toggle 0->1	Toggle 1->0
full	Yes	Yes	Yes
empty	Yes	Yes	Yes
x_full	Yes	Yes	Yes
w_full	Yes	Yes	Yes
state[2:0]	Yes	Yes	Yes
out_lat_count[1:0]	Yes	Yes	Yes
out_hit	Yes	Yes	Yes

Summary

Total Module Definition Coverage Summary

SCORE	LINE	TOGGLE
97.32	94.87	99.77

Total modules in report: 9

Expand All Collapse All

NAME	SCORE	LINE	TOGGLE
SystolicCtrl	96.88	93.75	100.00
fixed_point_combinational_Multiplie	99.17		99.17
SystolicPE	100.00	100.00	100.00
SyncFIFO	100.00	100.00	100.00
cmn_EnResetReg	100.00	100.00	100.00
SystolicDpath	100.00		100.00
SyncFIFO_ReadPtr	100.00	100.00	100.00
SystolicArray	100.00		100.00
SyncFIFO_WritePtr	100.00	100.00	100.00

Line coverage score for SystolicCtrl is due to untested default case statements within the output transition and latency counting logic. FP combinational multiplication on unsigned inputs were not tested.

Physical Design

Technical details beyond what is provided below cannot be shared under the non-disclosure agreement between Cornell Custom Silicon Systems, MPW (Multi-Project-Wafer) University Service (MUSE), and Taiwan Semiconductor Manufacturing Company (TSMC).

Using our TSMC 180 nm process node, an 8x8 array with FP Q4.4 numbers might be the most reasonable and practical configuration of our systolic array.

Synthesis

Area

```
*****
Report : area
Design : SystolicArray
Version: V-2023.12-SP5
Date   : Sun May  4 14:06:33 2025
*****
```

Library(s) Used:

Number of ports:	360
Number of nets:	20860
Number of cells:	19311
Number of combinational cells:	16732
Number of sequential cells:	2578
Number of macros/black boxes:	0
Number of buf/inv:	3531
Number of references:	29

```
Combinational area:      268396.126513
Buf/Inv area:           32938.975658
Noncombinational area:  109103.637085
Macro/Black Box area:   0.000000
Net Interconnect area:  undefined (Wire load has zero net area)
```

```
Total cell area:      377499.763597
Total area:           undefined
```

Hierarchical area distribution

	Global cell area		Local cell area			
Hierarchical cell	Absolute Total	Percent Total	Combi-national	Noncombi-national	Black-boxes	Design
SystolicArray	377499.7636	100.0	645.3888	757.3440	0.0000	SystolicArray
dpath	376097.0308	99.6	267750.7377	108346.2931	0.0000	SystolicDpath_SIZE8_NBITS8_DBITS4
Total			268396.1265	109103.6371	0.0000	

Timing (Worst-Case Setup)

```
*****
Report : timing
        -path full
        -delay max
        -max_paths 999
Design : SystolicArray
Version: V-2023.12-SP5
Date   : Sun May  4 14:06:32 2025
*****

# A fanout number of 1000 was used for high fanout net computations.

Operating Conditions: NCCOM   Library: 
Wire Load Model Mode: segmented

Startpoint: dpath/g_tensor_fifo_0_TensorFIFO_w_ptr_blk_w_ptr_reg_2_
            (rising edge-triggered flip-flop clocked by SystolicArray_clk)
Endpoint:   dpath/g_pe_rows_0_g_pe_cols_0_PE_sum_reg_q_reg_7_
            (rising edge-triggered flip-flop clocked by SystolicArray_clk)
Path Group: SystolicArray_clk
Path Type:  max

-----
data required time                4.94
data arrival time                 -4.89
-----
slack (MET)                       0.04
```

Place & Route

